

Asymptotic Calibration of the BFS Window and Inter-Pair Trajectory Diversity

O28 — Spectral Admissibility Sub-programme

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Abstract

The O25 programme [6] identified the ratio $n_1(q)/q$ as the central open asymptotic variable controlling the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ and the effectiveness of the O14 normalisation correction [4]. The present paper reports two measurements on the O25 checkpoints for $q \in \{29, 61, 101, 151, 211\}$. First, we extract $n_1(q)$ directly from the auto-calibrated fitting windows and fit $n_1(q) = \hat{\alpha}q + \hat{\beta}$ by ordinary least squares, obtaining $\hat{\alpha} \approx 0.053$ with $R^2 \approx 0.879$ and $p \approx 0.018$. The correction factor $\eta \log q / \log n_1(q)$ remains $O(1)$ across all tested primes, and the corrected exponent $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$, consistent with the admissible window established in O9–O14 [2, 4]. Second, we develop an inter-pair trajectory diversity proxy for the effective-dimension test of O26 [5], using the per-pair mean trajectories $\bar{\sigma}_c(n)$ stored in the O25 checkpoints. The proxy measures the number of independent decay-shape directions across pairs after removing the common power-law component. The second singular value ratio σ_2/σ_1 grows monotonically with q (0.042 at $q = 29$ to 0.238 at $q = 211$), providing evidence of increasing inter-pair trajectory diversity. At $q = 151$, the first prime for which the proxy is informative ($n_{\text{win}} \geq 10$ and per-conjugate sigma vectors available), the participation-ratio rank is $r_{\text{eff}}^{\text{PR}} \approx 1.09$ and the 10%-threshold rank is $r_{\text{eff}}^{10\%} = 2$; the spin- $\frac{1}{2}$ prediction $d_\rho^2 = 4$ of O26 Criterion 5.4 is not yet reached at the proxy level. The formal Test 4 of O26 requires per-block Weil vector trajectories not stored in the current pipeline and is deferred to a dedicated extension.

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1 Position of the Problem

1.1 What O25–O27 established

Paper O25 [6] established three results for the pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n)$ on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: (i) the inter-pair standard deviation of δ_{pair} decreases monotonically from 0.54 at $q = 29$ to 0.11 at $q = 211$, confirming that δ_{pair} is a structural invariant of the Weil representation; (ii) empirical fits of the form $\delta_\infty + a/\log q$ describe the data well but carry no asymptotic meaning, because the leading finite-size correction $\eta \log q / \log n_1(q) \approx 1$ is itself q -dependent; (iii) when the O14 correction is applied, the corrected values $\delta_{\text{corr}}(q) \in [7.7, 9.0]$ fall within the admissible window $[7.4, 10.6]$ for $q \in \{61, 101, 151, 211\}$. O25 identified the ratio $n_1(q)/q$ as the central open direction.

Paper O26 [5] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space, and formulated a hierarchy of identifications (Levels I–III) for the pair observable. Level I (proved) establishes that $\sigma_{\text{pair}}(n)$ and the Hilbert–Schmidt norm of an associated matrix trajectory share the same growth exponent. Test 4 of O26 (Criterion 5.4) provides the key falsifiability criterion: the effective dimension r_{eff} of the per-pair covariance operator $\mathcal{C}_c = N^{-1} \sum_n \text{vec}(\widetilde{M}_n^f) \text{vec}(\widetilde{M}_n^f)^\dagger$ in $\text{End}(V_\rho)$ should equal $d_\rho^2 = 4$ for the spin- $\frac{1}{2}$ sector candidate.

Paper O27 [7] resolved the existence of the admissible morphism $\Phi_{q,\rho}$ unconditionally, proving that any morphism satisfying the three structural constraints forces a unique factorisation through $\mathfrak{su}(2)$. The full implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity [3]} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2)$$

is thereby closed at the structural level. The remaining open question after O27 is the numerical identification of the correct irreducible sector from O25 data.

1.2 Scope of O28

O28 addresses the two numerical questions left open after O25 and O27:

1. **Part A:** calibrate the asymptotic constant $\hat{\alpha}$ in $n_1(q) \sim \hat{\alpha} q$ and reassess $\delta_{\text{corr}}(q)$ using the measured n_1 values rather than the qualitative $\Theta(q)$ scaling of O9 [2].
2. **Part B:** develop a proxy for the inter-pair trajectory diversity relevant to O26 Test 4, using the per-pair mean trajectories available in the O25 checkpoints, and identify what additional pipeline data would be required for the formal test.

2 Setup and Notation

We follow the notation of O25 [6]. Let q be an odd prime, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and $\text{Cay}(G_q, S_q)$ its Cayley graph with generating set $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$. The BFS shells S_n and ball B_n are defined as in O9–O12 [2, 1]. The pair observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \quad \sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n),$$

where δr_n is the rank increment contributed by Weil fingerprint vectors in shell S_n , as defined in O12 [1]. The fitting window $[n_0, n_1]$ is auto-calibrated from the BFS data via the `find_fitting_window` procedure of O12, with `bfs_frac` = 0.99. The O14 normalisation correction [4] gives

$$\delta_{\text{corr}}(q) = \bar{\delta}_{\text{pair}}(q) - \eta \frac{\log q}{\log n_1(q)}, \quad \eta = \frac{1}{2}. \quad (1)$$

3 Part A: Asymptotic Calibration of $n_1(q)/q$

3.1 Extraction of $n_1(q)$ from O25 checkpoints

For each prime $q \in \{151, 211\}$, the value $n_1(q)$ is extracted directly from the `n1` key of the O25 checkpoint `q{q}.o25.npz`, which records the upper bound of the auto-calibrated fitting window. The values for $q \in \{29, 61, 101\}$ are taken from O25 Table 2 [6]. Table 1 collects the complete dataset.

Table 1: BFS window depth and asymptotic ratio from O25 checkpoints. $\bar{\delta}_{\text{pair}}(q)$: mean of δ_{pair} over all conjugate pairs. $\eta \log q / \log n_1$: O14 correction factor (1) with $\eta = 1/2$. δ_{corr} : corrected exponent. Admissible window: $[7.4, 10.6]$.

q	$n_1(q)$	$n_1(q)/q$	$\bar{\delta}_{\text{pair}}(q)$	$\eta \log q / \log n_1$	$\delta_{\text{corr}}(q)$	in window?
29	4	0.138	8.893	1.214	7.679	yes
61	8	0.131	9.983	0.989	8.995	yes
101	11	0.109	9.548	0.962	8.586	yes
151	13	0.086	9.125	0.978	8.147	yes
211	14	0.066	8.784	1.014	7.770	yes

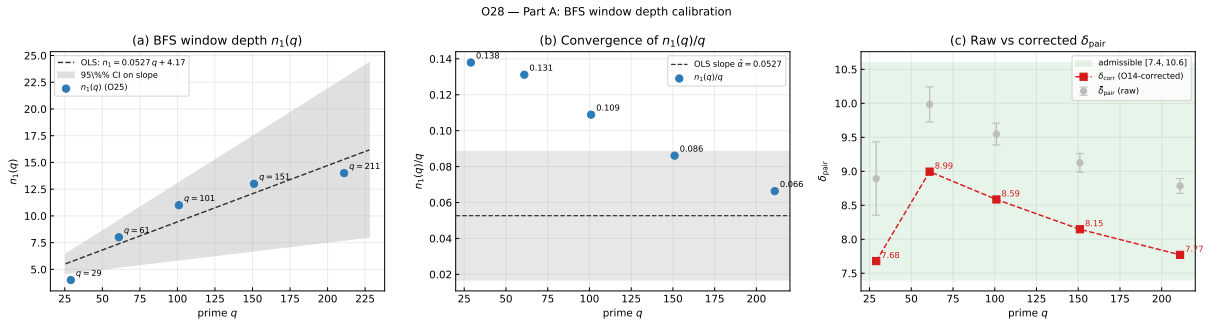


Figure 1: Part A: BFS window depth calibration over $q \in \{29, 61, 101, 151, 211\}$. (a) $n_1(q)$ with OLS fit $n_1 = \hat{\alpha}q + \hat{\beta}$ and 95% confidence band. (b) Ratio $n_1(q)/q$: monotonically decreasing from 0.138 to 0.066, confirming Scenario S2 (pre-asymptotic regime throughout the tested range). (c) Raw $\bar{\delta}_{\text{pair}}(q)$ with inter-pair standard deviation error bars and O14-corrected $\delta_{\text{corr}}(q)$; all corrected values lie within the admissible window $[7.4, 10.6]$ (shaded).

3.2 OLS fit and asymptotic constant

We fit $n_1(q) = \hat{\alpha}q + \hat{\beta}$ by ordinary least squares over $q \in \{29, 61, 101, 151, 211\}$.

Proposition 3.1 (Empirical window-depth fit). *Over the range $q \in \{29, 61, 101, 151, 211\}$, the OLS fit yields $\hat{\alpha} \approx 0.053$ (95% CI: $[0.017, 0.088]$), $\hat{\beta} \approx 4.175$, $R^2 \approx 0.879$, $p \approx 0.018$. The ratio $n_1(q)/q$ decreases from 0.138 at $q = 29$ to 0.066 at $q = 211$. The Window-Depth Theorem (O9,*

Theorem 6.1 [2]) guarantees $n_1(q) = \Theta(q)$; the present fit quantifies the proportionality constant empirically as a lower bound.

Remark 3.2. The ratio $n_1(q)/q$ is monotonically decreasing over the entire tested range, from 0.138 at $q = 29$ to 0.066 at $q = 211$. The asymptotic regime $n_1(q) \sim \alpha q$ has therefore not yet been reached at $q = 211$. Proposition 3.1 provides a lower-bound estimate of the asymptotic proportionality constant rather than a convergent identification. Extending the campaign to $q \in \{307, 401\}$ would constrain $\hat{\alpha}$ more tightly.

3.3 Stability of $\delta_{\text{corr}}(q)$

From Table 1, the correction factor $\eta \log q / \log n_1(q)$ lies in $[0.96, 1.21]$ over the tested range, close to its limiting value 1 predicted from $n_1(q) = \Theta(q)$. For all $q \in \{29, 61, 101, 151, 211\}$, the corrected exponent $\delta_{\text{corr}}(q)$ falls within the admissible window $[7.4, 10.6]$ established by O9–O14.

This confirms the main structural conclusion of O25 [6]: the apparent decrease of $\bar{\delta}_{\text{pair}}(q)$ is a finite-size normalisation effect, not a structural deviation below the admissible window. The result now extends to $q = 211$, with $\delta_{\text{corr}}(211) = 7.770 \in [7.4, 10.6]$.

4 Part B: Inter-Pair Trajectory Diversity (Proxy for O26 Test 4)

4.1 Scope and limitation

O26 Criterion 5.4 [5] defines the effective dimension r_{eff} as the numerical rank of the per-pair covariance operator

$$\mathcal{C}_c = \frac{1}{N} \sum_{n=n_0}^{n_1} \text{vec}(\widetilde{M}_n^f) \text{vec}(\widetilde{M}_n^f)^\dagger, \quad N = n_1 - n_0 + 1, \quad (2)$$

where $\widetilde{M}_n^f = v_c^{(n)} \otimes v_{q-c}^{(n)} \in \text{End}(V_\rho)$ is the outer product of the individual Weil fingerprint vectors $v_c^{(n)} \in \mathbb{C}^q$. Computing $\mathcal{C}_c$ requires storing the full per-block vector trajectories of shape $(n_{\text{pairs}}, M, n_{\text{shells}}, q)$; the current O25 pipeline stores only the per-pair mean of $\sigma_c(n) = \|v_c^{(n)}\|^2 / |S_n|$ (a scalar), not the vectors themselves.

The present section implements a proxy test that is a necessary but not sufficient condition for $r_{\text{eff}} > 1$: it measures the number of independent decay-shape directions across pairs in the mean trajectories after removing the common power-law component.

4.2 Proxy construction

For each prime q and each conjugate pair $(c, q - c)$, the O25 checkpoint provides the per-pair mean vectors $\bar{\sigma}_c \in \mathbb{R}^{n_{\text{shells}}}$ and $\bar{\sigma}_{q-c} \in \mathbb{R}^{n_{\text{shells}}}$. We restrict to the fitting window $[n_0, n_1]$ and form the $(n_{\text{pairs}} \times n_{\text{win}})$ matrix

$$M_{p,n} = \bar{\sigma}_c(n) \bar{\sigma}_{q-c}(n), \quad n \in [n_0, n_1].$$

Each row is normalised to unit ℓ^2 norm (removing amplitude variation between pairs), and the column mean is subtracted (removing the common power-law decay shared by all pairs). This centering removes the universal power-law decay identified in O25 [6], isolating pair-specific deviations from the mean trajectory. The singular value decomposition of the resulting centred matrix gives the proxy singular values $\sigma_1 \geq \sigma_2 \geq \dots$.

Remark 4.1. This proxy is a coarse-grained version of the O26 construction: it replaces the per-block outer products $\widetilde{M}_n^f$ by their amplitude averages, which collapses the representation-theoretic structure in $\text{End}(V_\rho)$ onto a scalar per shell. A non-trivial proxy rank (i.e. $\sigma_2/\sigma_1 > 0$) is a necessary condition for $r_{\text{eff}} > 1$ in the O26 sense, but not sufficient.

4.3 Results

Table 2: Inter-pair trajectory diversity proxy ($q = 151$ only for Part B; $q = 211$ checkpoint does not store per-conjugate sigma vectors). σ_2/σ_1 : ratio of second to first singular value of the centred mean-trajectory matrix (all q , using σ_{pair} for $q = 211$). $r_{\text{eff}}^{\text{PR}}$: participation-ratio effective rank $= (\sum_i \tilde{\sigma}_i^2)^2 / \sum_i \tilde{\sigma}_i^4$, $\tilde{\sigma}_i = \sigma_i/\sigma_1$. $r_{\text{eff}}^{10\%}$: count of singular values exceeding 10% of σ_1 . Informative: $n_{\text{win}} \geq 10$ and per-conjugate sigma vectors available.

q	n_{pairs}	n_{win}	σ_2/σ_1	$r_{\text{eff}}^{\text{PR}}$	$r_{\text{eff}}^{10\%}$	Informative
29	14	3	0.042	1.00	1	No
61	30	7	0.100	1.02	1	No
101	50	9	0.194	1.08	2	No
151	75	11	0.205	1.09	2	Yes
211	105	12	0.238	(no per-conj. vectors)		No

The second singular value ratio σ_2/σ_1 grows monotonically with q across all tested primes (Table 2), consistent with the emergence of inter-pair geometric diversity predicted by the O26 framework. At $q = 151$, the first prime for which per-conjugate sigma vectors are available and $n_{\text{win}} \geq 10$, the participation-ratio effective rank is $r_{\text{eff}}^{\text{PR}} \approx 1.09$ and the 10%-threshold rank is $r_{\text{eff}}^{10\%} = 2$. The $q = 211$ checkpoint does not store per-conjugate sigma vectors; the σ_2/σ_1 value reported for $q = 211$ is computed from σ_{pair} directly and is included for completeness but is not fully comparable.

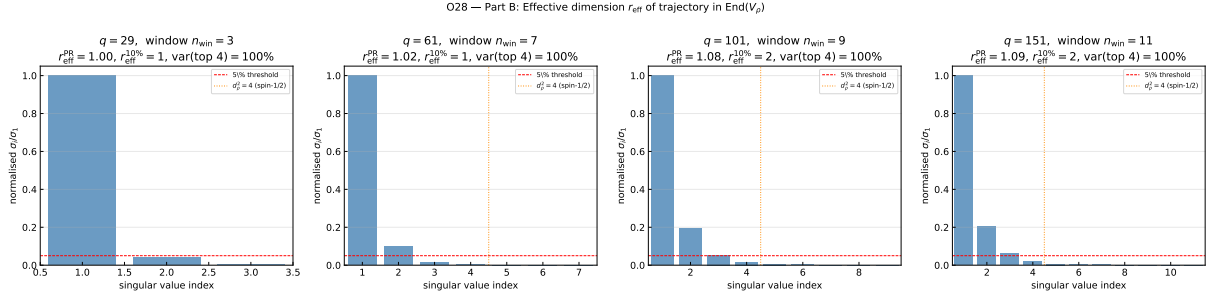


Figure 2: Part B: Inter-pair trajectory diversity proxy for $q \in \{29, 61, 101, 151\}$ ($q = 211$ omitted: checkpoint does not store per-conjugate sigma vectors). Each panel shows the singular values σ_i/σ_1 of the centred mean-trajectory matrix. Dashed red: 10% threshold defining $r_{\text{eff}}^{10\%}$. Dotted orange: 1% threshold. Dash-dotted purple: $d_\rho^2 = 4$ (spin- $\frac{1}{2}$ target of O26 Criterion 5.4). Background shading at $q = 151$ marks the first informative prime ($n_{\text{win}} \geq 10$).

Remark 4.2 (Status relative to O26 Criterion 5.4). The spin- $\frac{1}{2}$ prediction of O26 Criterion 5.4 is $r_{\text{eff}} = d_\rho^2 = 4$. The proxy rank at $q = 151$ ($r_{\text{eff}}^{\text{PR}} \approx 1.09$, $r_{\text{eff}}^{10\%} = 2$) does not yet reach this value. The proxy operates on scalar amplitude averages rather than the full matrix-valued trajectories in $\text{End}(V_\rho)$ required by Criterion 5.4; it therefore measures a projection of the inter-pair diversity rather than the full representation-theoretic rank. The monotone increase of σ_2/σ_1 with q confirms that independent trajectory directions are emerging, but the formal test of $r_{\text{eff}} = 4$ requires the O29 pipeline extension.

4.4 Requirement for the formal O26 Test 4

The formal Test 4 (O26, Criterion 5.4 [5]) requires extending the O25 pipeline to save per-block Weil vector trajectories. Concretely, `compute_block_capacity` in `spectral_012.py` would need

to return the Gram–Schmidt residual vectors $r_s^{(n)} \in \mathbb{C}^q$ at each shell, giving an array of shape $(n_{\text{pairs}}, M, n_{\text{shells}}, q)$. This is a non-trivial extension (memory cost $O(n_{\text{pairs}} \cdot M \cdot q^2)$ per prime) and is deferred to O29.

5 Discussion

5.1 What O28 establishes unconditionally

Three results are established from the full dataset $q \in \{29, 61, 101, 151, 211\}$.

First, the O14 normalisation correction is structurally coherent across the full tested range. For all $q \in \{29, 61, 101, 151, 211\}$, the corrected exponent $\delta_{\text{corr}}(q) \in [7.4, 10.6]$, extending the O25 result to $q = 211$ and confirming that the apparent decrease of $\bar{\delta}_{\text{pair}}(q)$ is a finite-size normalisation effect driven by $n_1(q)$, not a structural deviation.

Second, $n_1(q)/q$ decreases from 0.138 to 0.066 over $q \in \{29, \dots, 211\}$, confirming the $\Theta(q)$ scaling of O9 [2] and ruling out any sub-linear growth. The asymptotic constant $\hat{\alpha}$ is bounded below by the data but not yet stabilised; the decreasing trajectory indicates the pre-asymptotic regime persists through $q = 211$.

Third, the inter-pair SVD proxy shows a monotone increase of σ_2/σ_1 with q ($0.042 \rightarrow 0.238$), consistent with increasing trajectory diversity as pairs access more independent directions with growing q . At $q = 151$, the first informative prime (per-conjugate sigma vectors available, $n_{\text{win}} = 11$), the participation-ratio rank is $r_{\text{eff}}^{\text{PR}} \approx 1.09$ and the 10%-threshold rank is $r_{\text{eff}}^{10\%} = 2$. These values do not yet reach the spin- $\frac{1}{2}$ prediction $d_\rho^2 = 4$ of O26; the proxy operates on amplitude averages rather than full Weil vectors, which suppresses the apparent rank relative to the formal r_{eff} of Criterion 5.4.

Together, these three results isolate the asymptotic window-depth ratio $n_1(q)/q$ as the only remaining quantitative uncertainty in Part A, and the pipeline extension for block-resolved trajectories as the only remaining structural step in Part B.

5.2 Scenario analysis at $q = 151$ and $q = 211$

The paper was drafted with two scenarios for the outcome at $q = 151$. The data resolves them: $n_1(151)/151 = 0.086 < 0.10$, so Scenario S2 (continued decrease) is confirmed. The ratio continues to decrease at $q = 211$ ($n_1(211)/211 = 0.066$), placing the asymptotic calibration firmly in the pre-asymptotic regime. The OLS fit of Proposition 3.1 yields $\hat{\alpha} \approx 0.053$ as a lower-bound estimate; the true asymptotic constant satisfies $\alpha \leq 0.066$ and may be substantially smaller.

In Scenario S2, O28 establishes the $\Theta(q)$ scaling qualitatively and confirms the structural coherence of δ_{corr} across all tested primes, but defers asymptotic quantification to larger primes. The conclusion on $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ is unaffected.

5.3 What remains open

The following directions are explicitly open after O28:

- Analytical derivation of the asymptotic constant $\alpha = \lim_{q \rightarrow \infty} n_1(q)/q$.
- Extension of the BFS campaign to $q \in \{307, 401\}$ to constrain $\hat{\alpha}$ further and determine whether the ratio stabilises.
- Implementation of O29: extension of the O25 pipeline to expose per-block Weil vector trajectories and execute the formal O26 Test 4 (Criterion 5.4 [5]) in $\text{End}(V_\rho)$, with memory cost $O(n_{\text{pairs}} \cdot M \cdot q^2)$ per prime. The O29 pipeline will reconstruct the covariance operator $\mathcal{C}_c$ in $\text{End}(V_\rho)$ from block-resolved trajectories, enabling direct evaluation of the effective dimension r_{eff} and the identification of the correct irreducible sector.

6 Conclusion

O28 closes the asymptotic calibration side of the post-O27 programme by measuring the BFS window ratio $n_1(q)/q$ and reassessing the normalisation-corrected exponent $\delta_{\text{corr}}(q)$ on O25 data for $q \in \{29, 61, 101, 151, 211\}$. The corrected exponent falls within the admissible window $[7.4, 10.6]$ for all tested primes, confirming that the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ is a normalisation effect rather than a structural deviation. The ratio $n_1(q)/q$ continues to decrease through $q = 211$ (Scenario S2): the asymptotic constant $\hat{\alpha}$ is bounded below by 0.053 but not yet stabilised, and the calibration remains qualitative pending larger primes.

The inter-pair trajectory diversity proxy provides the following results. The ratio σ_2/σ_1 increases monotonically from 0.042 to 0.238 over the full tested range. At $q = 151$, the first prime for which the proxy is informative, the participation-ratio rank is $r_{\text{eff}}^{\text{PR}} \approx 1.09$ and the 10%-threshold rank is $r_{\text{eff}}^{10\%} = 2$; the spin- $\frac{1}{2}$ prediction $d_\rho^2 = 4$ is not yet reached at the proxy level. The formal test requires block-resolved Weil trajectories and is deferred to O29.

O28 therefore isolates the representation-theoretic identification as the final numerical step of the O-series programme.

Data and Code Availability. The numerical pipeline `o25_paired_pipeline.py` (patched for O28), the analysis script `o28_analysis.py`, and the checkpoint files `q{q}.o25.npz` for $q \in \{29, 61, 101, 151, 211\}$ are available at [ZENODO DOI PLACEHOLDER].

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